

Harsh Garg

Punjab, India | +91 9041594669 | gargharsh2307@gmail.com

GitHub: github.com/Harshgg1 | LinkedIn: linkedin.com/in/harsh-garg-71a465328/

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, C

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSockets

Databases: PostgreSQL, Redis, MongoDB, Prisma ORM

DevOps: Docker, Docker Compose, Git, GitHub, Turborepo

Core CS: Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

PROJECTS

Distributed Real-Time Chat Platform

- Engineered a production-style real-time messaging platform using **Next.js, Node.js, Express, PostgreSQL, Prisma, Redis Pub/Sub, WebSockets and Docker Compose**.
- Implemented **JWT authentication**, bcrypt password hashing and protected REST APIs to secure user sessions and WebSocket communication.
- Designed persistent room-based messaging with PostgreSQL supporting pagination, message editing/deletion, typing indicators, read receipts and presence tracking.
- Leveraged **Redis Pub/Sub** to synchronize messages across backend instances, enabling horizontally scalable real-time communication.
- Built automatic WebSocket reconnection, room management and modular backend architecture for maintainability.

CourseMint – Full Stack Course Management Platform

- Built a full-stack course management platform using **React, Node.js, Express and MongoDB** with responsive UI and REST APIs.
- Developed modular backend services for authentication, course management and business logic following scalable routing architecture.
- Designed reusable frontend components and integrated persistent data storage for users and courses.
- **Currently migrating** the platform to PostgreSQL, Redis caching and Docker Compose for improved scalability.

ACHIEVEMENTS

- Strong foundation in Data Structures & Algorithms with consistent problem-solving practice.
- Focused on backend engineering, distributed systems, scalable application design and modern web technologies.

EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering in Computer Engineering (2024–2028)

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks